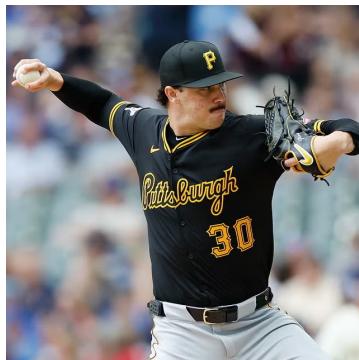

When Should You Pull a Starting Pitcher?

— A Data-Driven Approach Using
Retrosheet Data —



Problem Context

- Starting Pitcher vs. Relief Pitcher



- Role of the Manager



Why Does This Matter?

- Pitching decisions impact game outcomes
- Tradeoff: Fatigue vs. Bullpen Cost
- Widely debated in baseball analytics

Research Question

When should a starting pitcher be removed from the game?

- Compare “stay” vs. “pull” decisions
- Evaluate real managerial decisions



Approach Overview

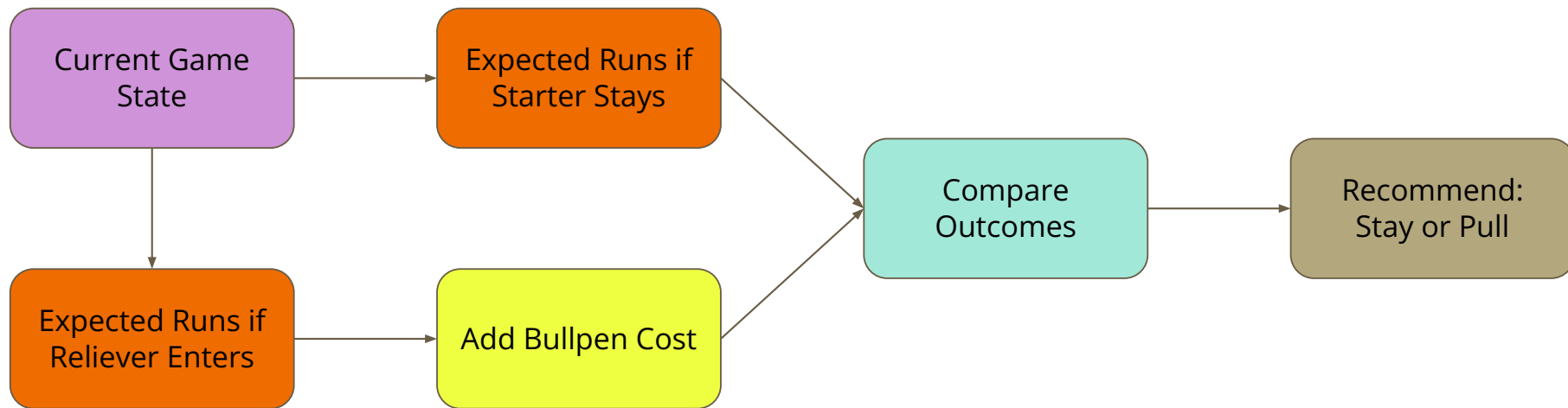
1. Build Run Expectancy model
2. Model pitcher fatigue and performance
3. Simulate “stay vs. pull” decisions
4. Compare model vs. managers

Data

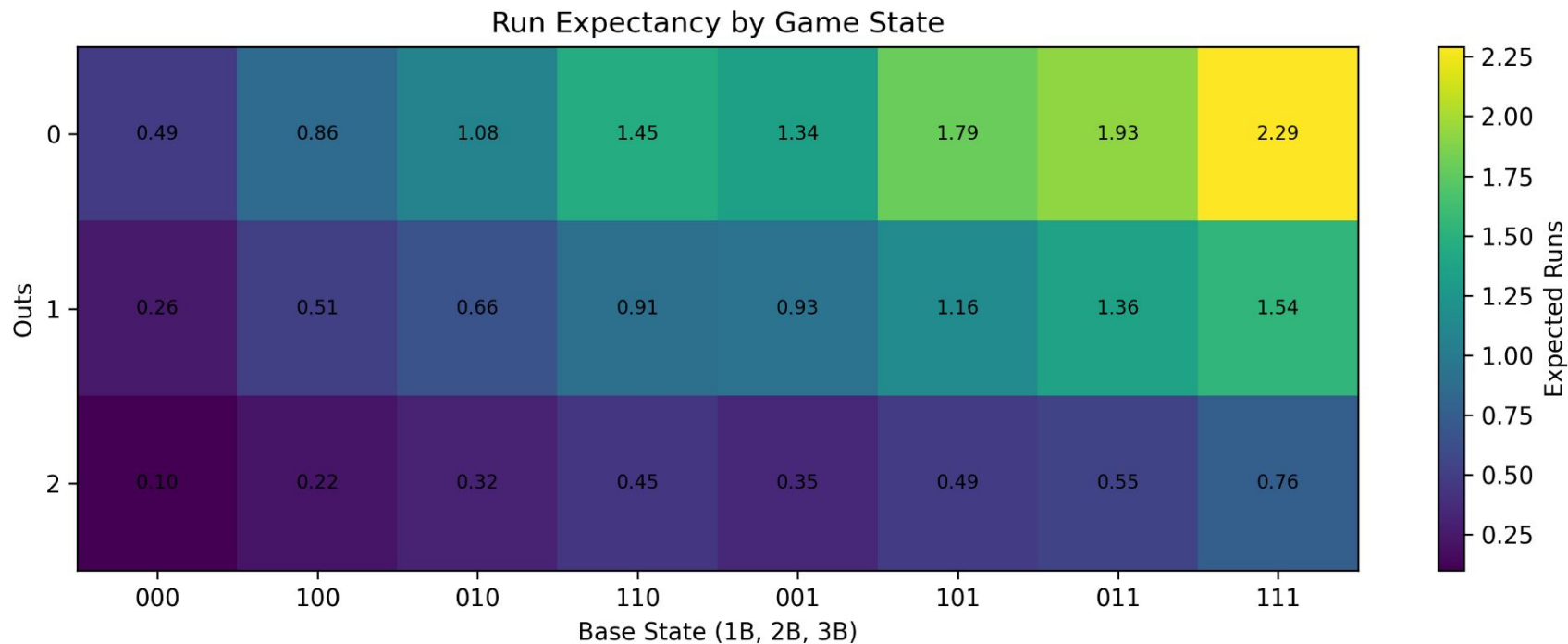
- Retrosheet event-level data (2022-2025)
 - Chadwick Tools (event files → csv files)
- Plate appearance level (~400K+ observations)
- Includes game state, players, outcomes

Modeling

Decision Framework: Stay vs. Pull



Run Expectancy Framework



Modeling Pitcher Fatigue

- # of pitches thrown
- Times through the order (TTO)
- Starter vs. Reliever

Incorporating In-Game Performance

- Runs allowed
- Hits/Walks allowed
- Recent performance (last 3 batters)

Modeling Expected Runs

Target:

- Expected Runs from Current State

Features:

- Game State: outs, base runners
- Pitcher Fatigue: batters faced, times through the order
- In-game Performance: runs allowed, recent performance
- Game Context: inning, score differential

$$\mathbf{E}[\text{runs}] = \mathbf{f}(\text{state}, \text{fatigue}, \text{performance}, \text{context})$$

Key Drivers of Expected Runs

Game State

These are the largest effects

- Bases loaded: **+1.087 runs**
- Runners on 2nd and 3rd: **+0.854 runs**
- Each out: **-0.289 runs**

Pitcher Fatigue

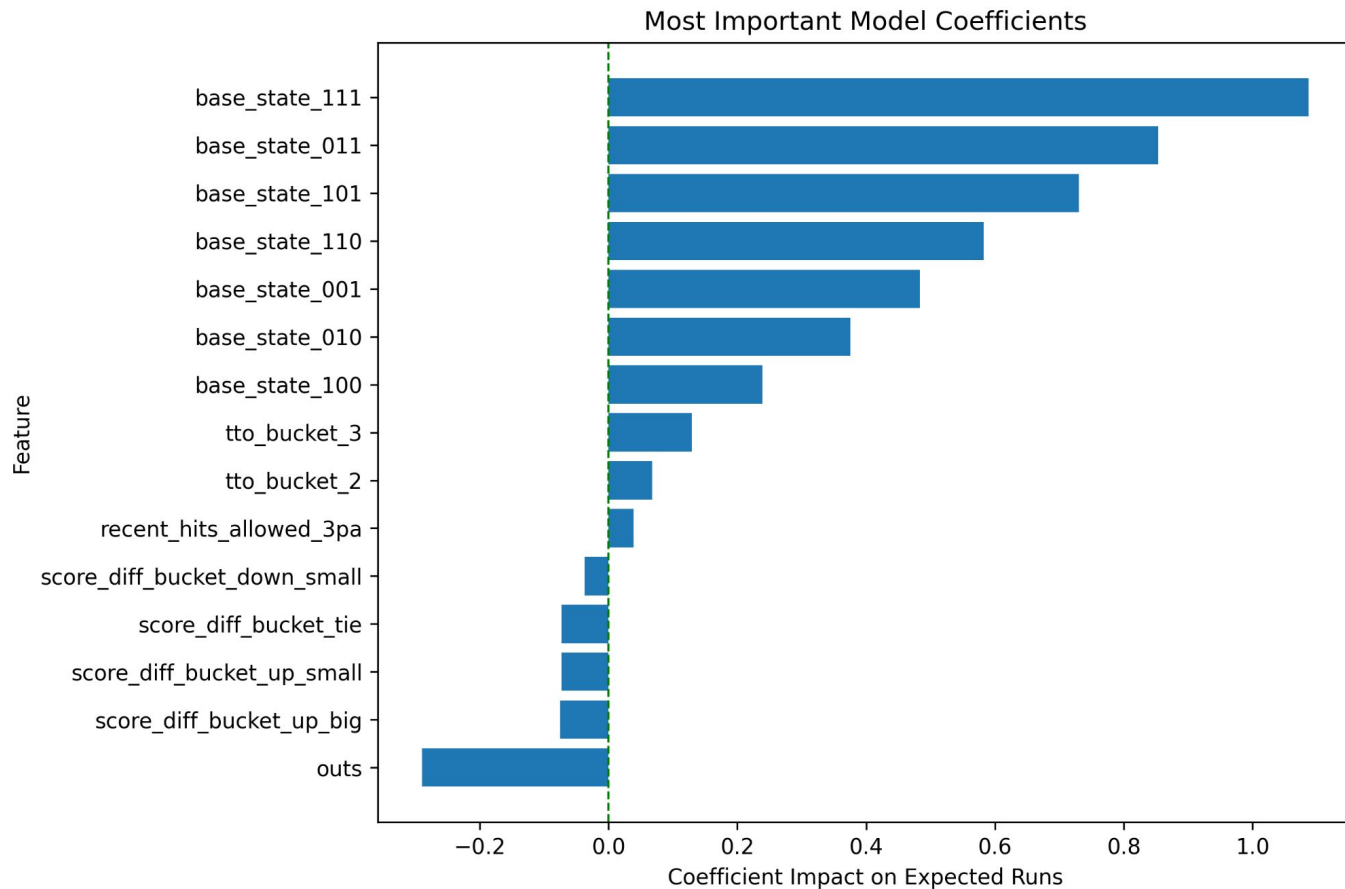
- 3rd time through the order: **+0.129 runs**
- 2nd time through: **+0.068 runs**

In-Game Performance

- Runs allowed so far: **+0.002 runs**
- Hits allowed last 3 PA: **+0.039 runs**
- Walks/HBP allowed last 3 PA: **+0.011 runs**
- Runs allowed last 3 PA: **-0.026 runs**
- Recent performance: **strong short-term signal**

Game Context

- Later innings → fewer runs remaining
- Close games behave differently than blowouts
 - Pitching team down small (1-3 runs): **-0.037 runs**
 - Tie game: **-0.072 runs**
 - Pitching team up small: **-0.072 runs**
 - Pitching team up big (4+ runs): **-0.075 runs**



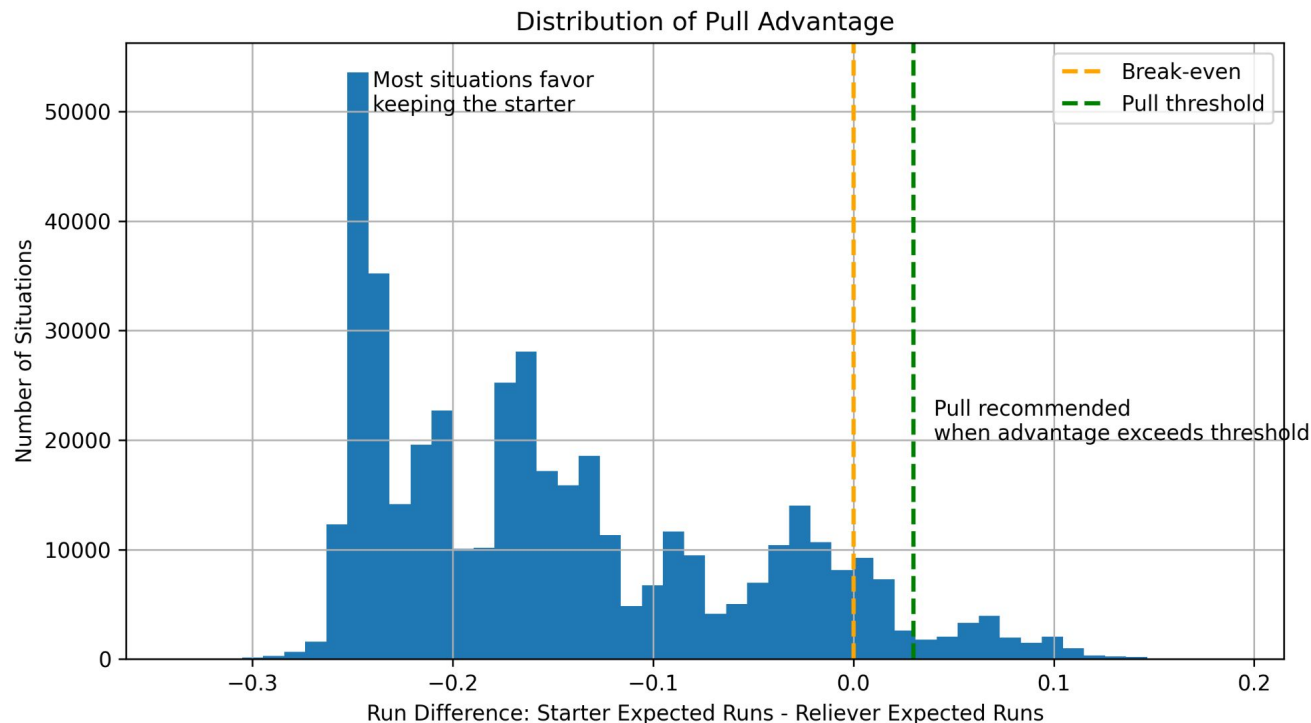
Game state dominates, while fatigue and performance provide additional context

Modeling the Bullpen

- Future bullpen cost (remaining innings)
- Reliever quality varies by inning
- Fresh reliever assumption



When Does the Model Recommend Pulling?

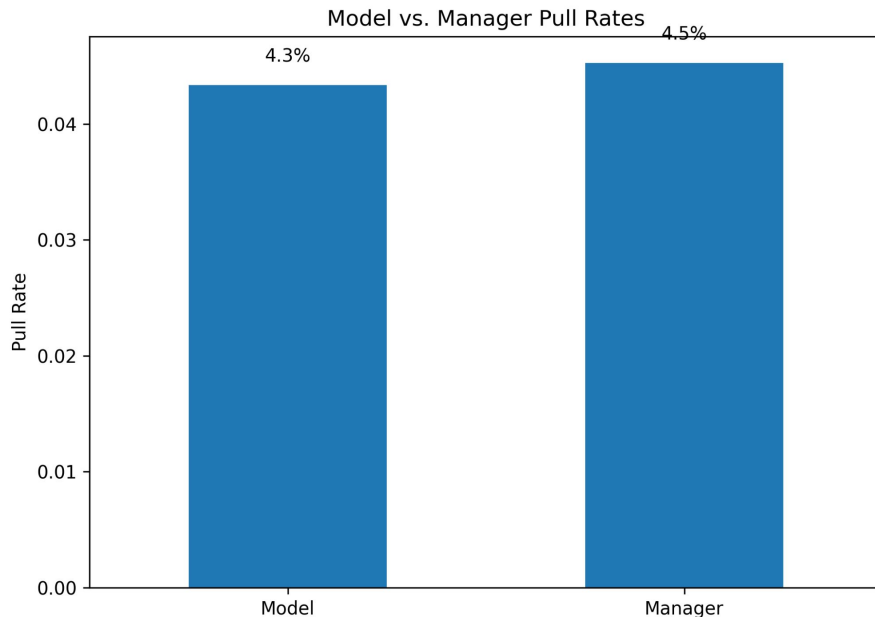


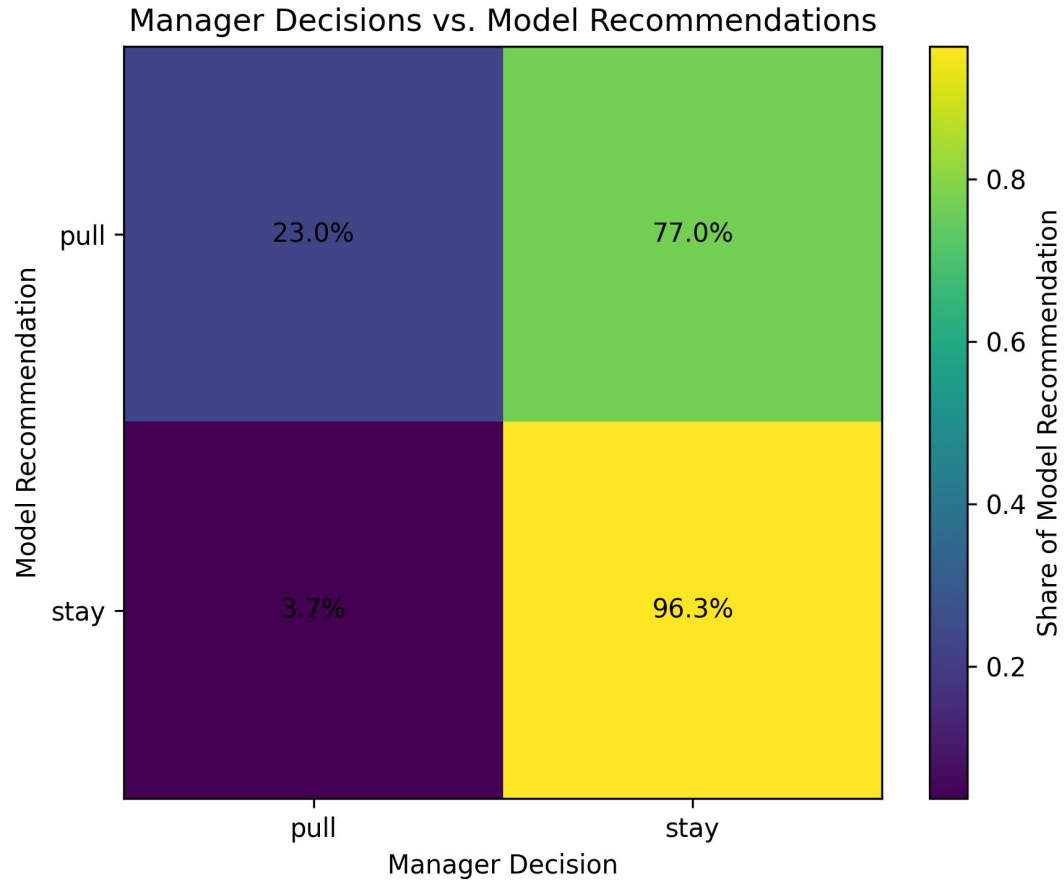
Most situations favor keeping the starter, but a small subset strongly favors pulling

Modeling Results

Overall Results

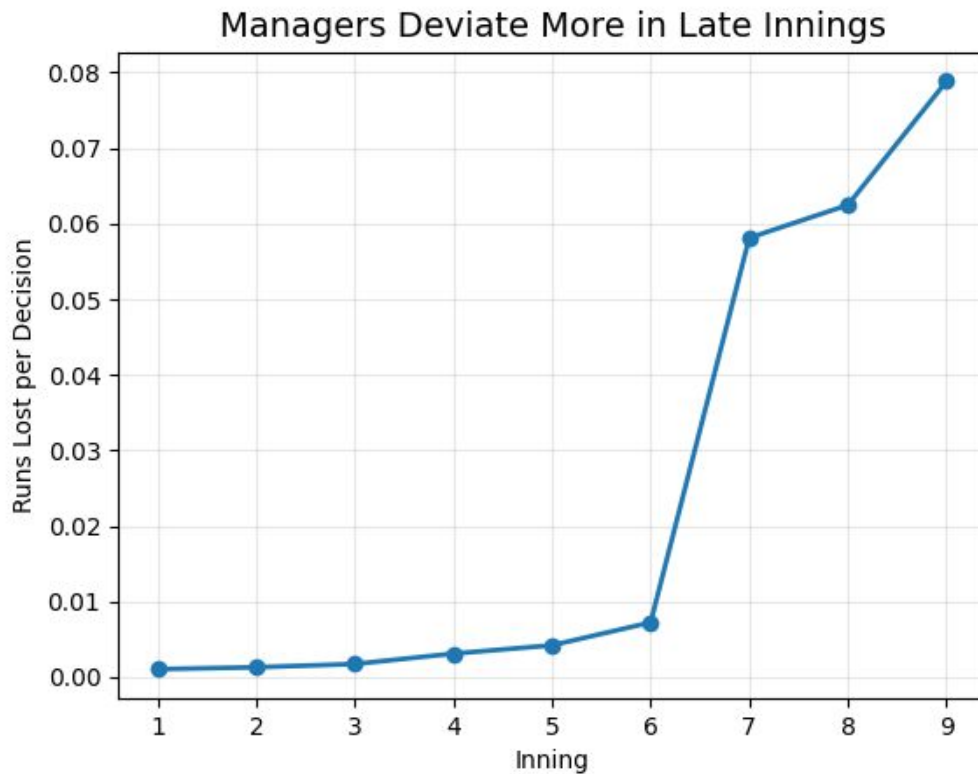
- Model vs manager pull rates
- Alignment rate (~93%)
- Average decision cost (~0.0046 runs)
- Managers are generally close to optimal, but small inefficiencies remain



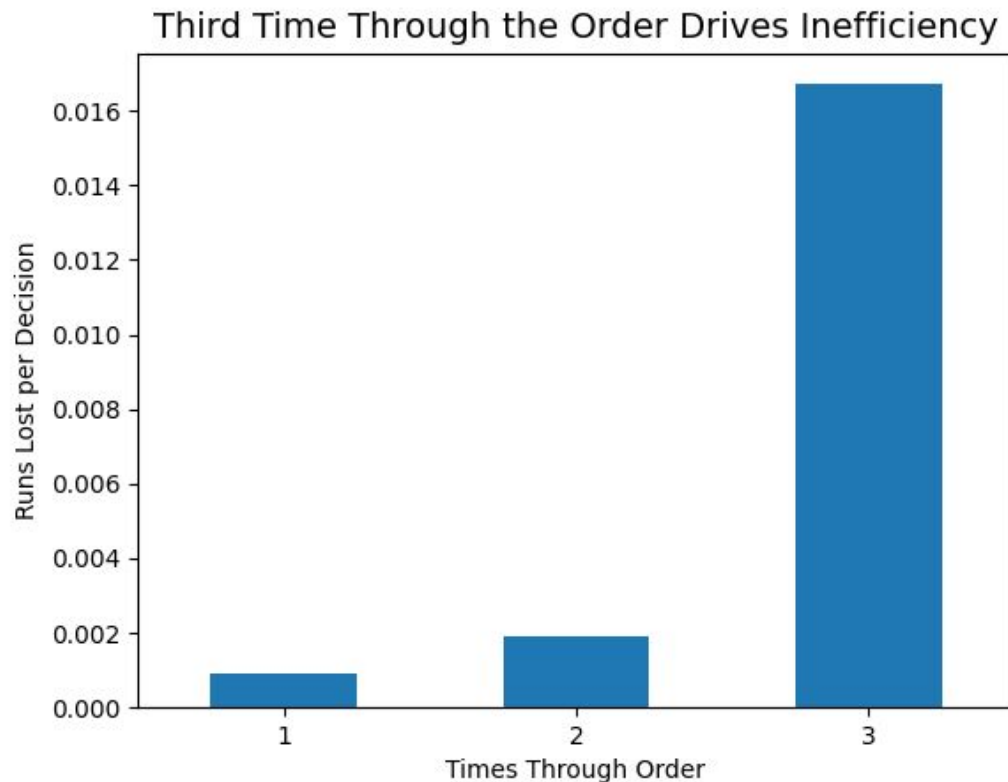


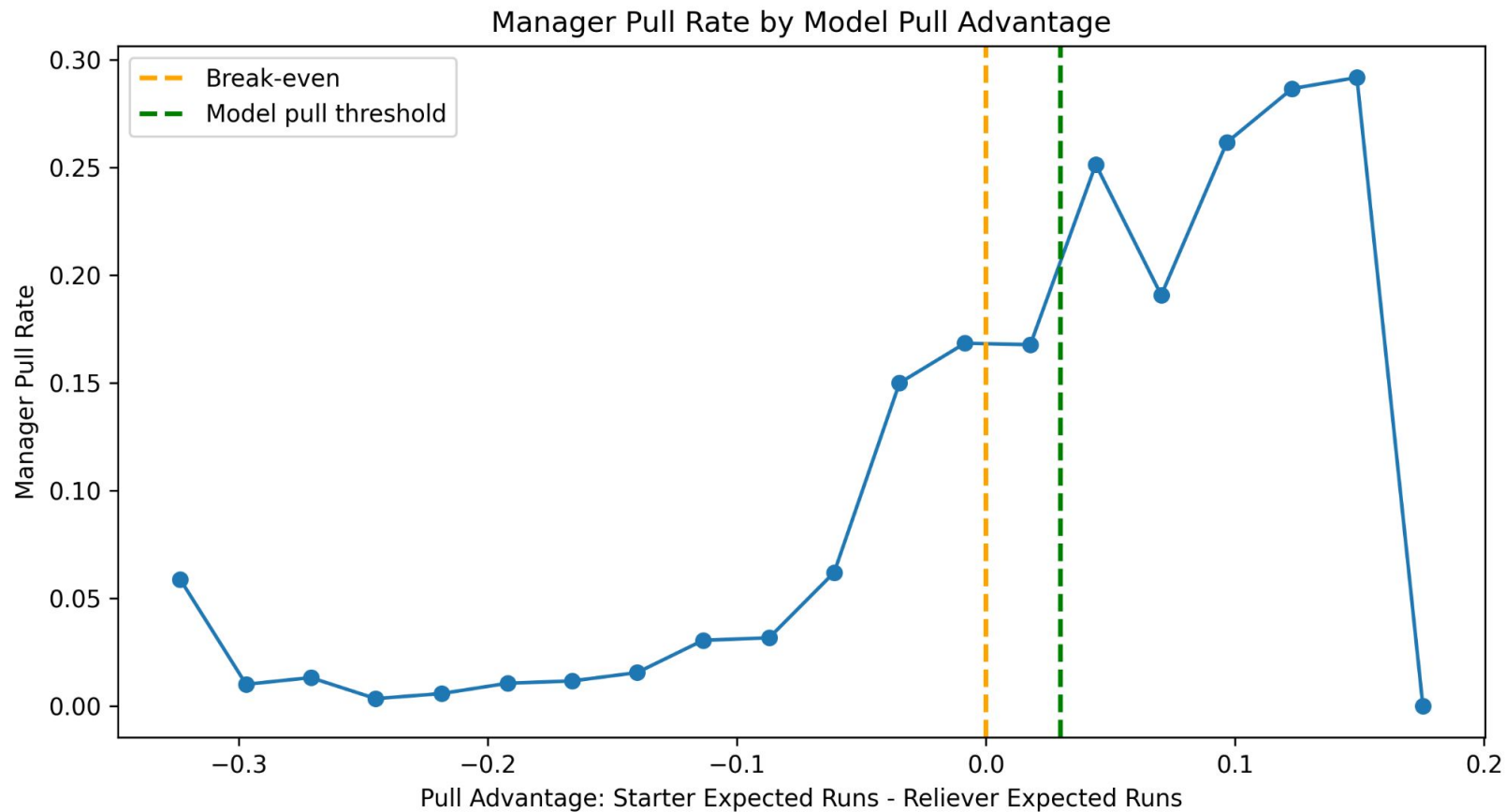
Where Do Mistakes Occur?

Where Do Managers Deviate?



Third Time Through the Order





Managers respond to advantage, but less aggressively than the model

Interpretation

- Managers are generally close to optimal
- Slight tendency to leave pitchers in too long
- Largest inefficiencies occur:
 - Late innings
 - Third time through the order

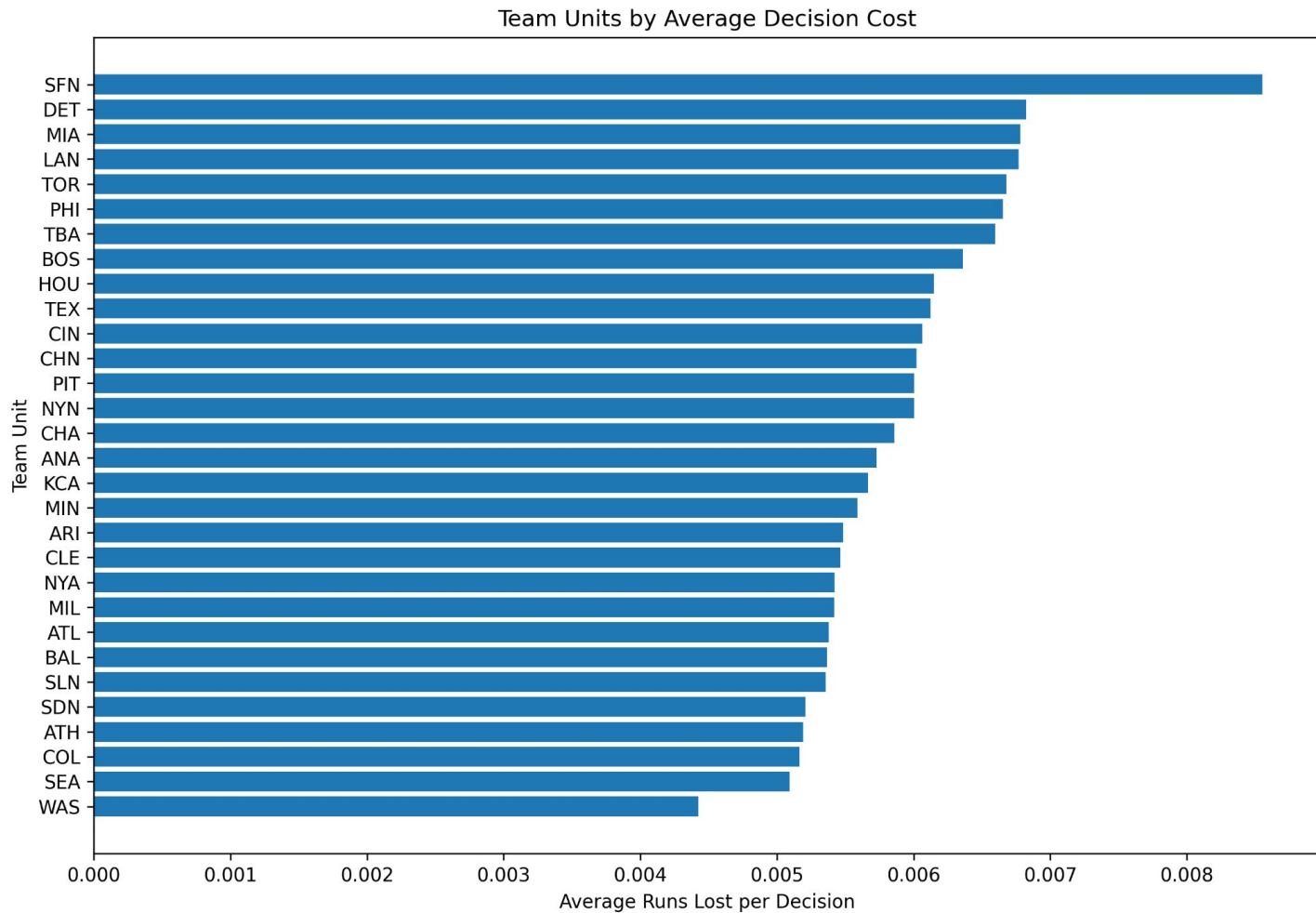
Model Assumptions

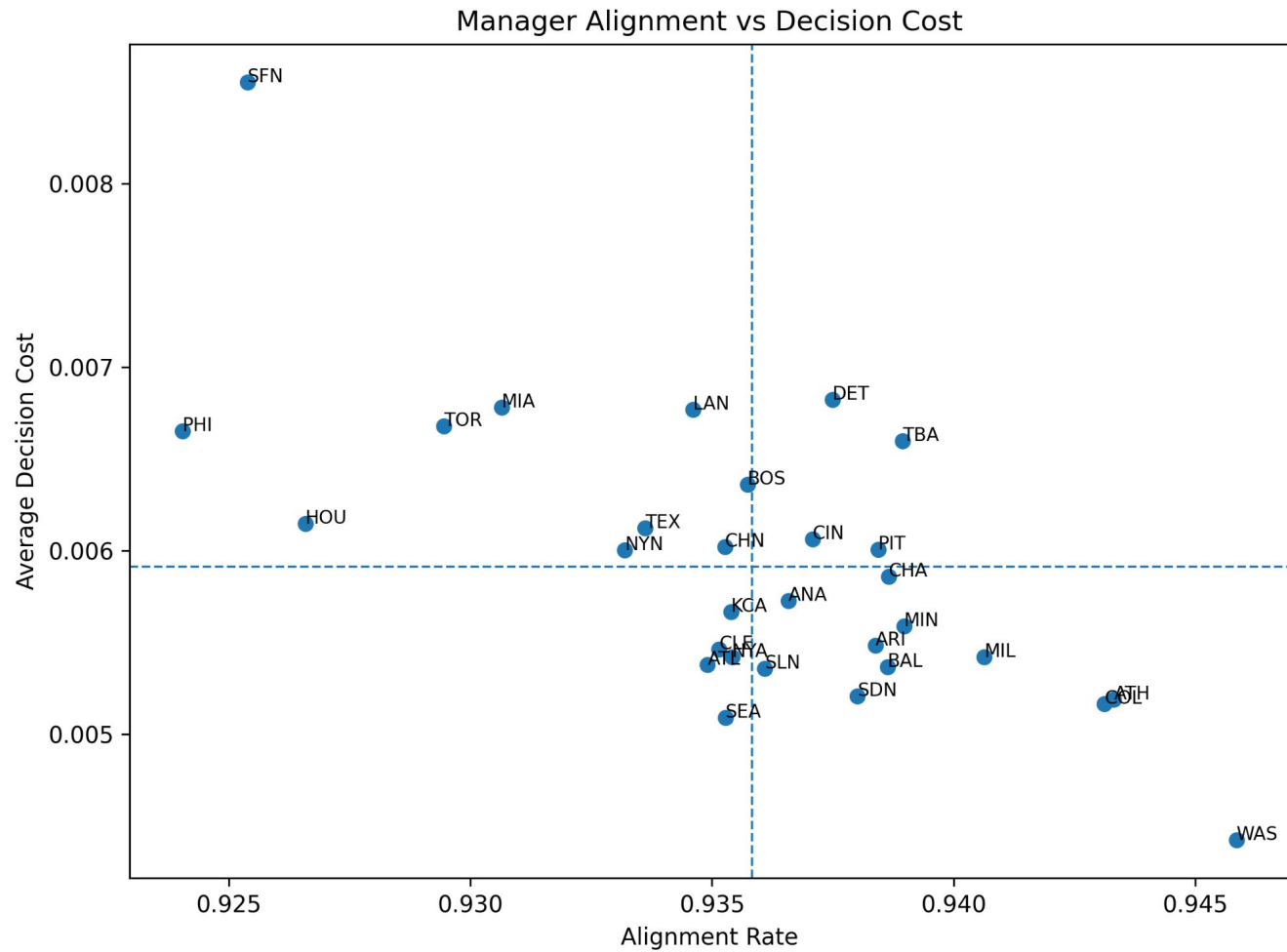
- Fresh reliever replaces starter
- Average bullpen quality
 - Inning-based reliever quality adjustment
- No strategic reliever sequencing

Limitations

- Does not model reliever sequencing
- Uses average reliever assumptions
- No team-specific bullpen strength
- No bullpen availability factor

MLB Results





Conclusion

Managers are close to optimal, but tend to leave starters in slightly too long (especially late in games and third time through)

- Alignment with the model is high, indicating strong overall decision-making
- Largest inefficiencies occur in high-leverage situations
- Teams that pull pitchers more aggressively tend to perform closer to optimal

Implications

- Small inefficiencies can accumulate into meaningful run impact
- Data-driven frameworks can improve decision-making
- Future work can incorporate reliever-specific modeling and bullpen strategy

References

Retrosheet. (2025). *Retrosheet event files (2022–2025)*.

<https://www.retrosheet.org/>

MLB pitching change example video.

Retrieved from <https://www.youtube.com/watch?v=w-35ghwAw90>